Software Engineer - ML/Computer Vision (Battery Sorting)

Redwood Materials · McCarran, NV; San Francisco, California, United States

Job ID (this archive)	ENERGY-099
Employer	Redwood Materials
Location(s)	McCarran, NV; San Francisco, California, United States
Career stage (archive tag)	Entry or experienced (Entry / Mid-career)
Location group	Other U.S.
Majors (archive tags)	Electrical Engineering, Computer Science, Software Engineering, Data Science
Energy area	Battery recycling and storage / data
Pay listed	Not listed
Recruiting status when checked	Open when checked: the employer page had a live application. Checked Oct 26, 2026.
Source page	https://job-boards.greenhouse.io/redwoodmaterials/jobs/6099577004?gh_jid=6099577004
Original link (from the course sheet)	https://boards.greenhouse.io/redwoodmaterials/jobs/6099577004?gh_jid=6099577004
Captured	2026-10-26
Course sheet refs	Expanded archive ENERGY-070

> Saved on 2026-10-26 from the employer's own careers page. The posting text below is the employer's own wording, reformatted for printing. Live listings change and close; check the source page before applying.

Job description

About Redwood Materials

Redwood Materials was founded in 2017 by JB Straubel, co-founder and former CTO of Tesla, on a simple but radical idea: the critical minerals powering the energy transition already exist. They just need to come back. Today Redwood is the largest lithium-ion battery recycler in North America — recovering, refining, and redeploying critical minerals from end-of-life batteries back into the supply chain. On the energy side, Redwood has deployed the largest second-life battery project in the world, delivering power to AI data centers in months rather than years using proprietary energy architecture engineered entirely in-house. The company is profitable, the market it operates in is projected to grow 20 to 50 times over the next decade, and Redwood is the only company with the supply chain, the technology, and the operational scale to lead it.

Software Engineer, ML/Computer Vision (Battery Sorting)

The Battery Sorting team at Redwood Materials is building a world-class, ML-enabled sorting platform that uses computer vision and machine learning to classify and route thousands of end-of-life batteries per hour across diverse chemistries and form factors. This role sits at the intersection of software engineering and machine learning, with direct ownership of the production systems powering automated battery sorting on the factory floor. The ideal candidate is equally comfortable debugging a production incident as iterating on a model, and will have the opportunity to generate patents in automated battery classification. This is a high-impact, highly visible role with immediate real-world application in advancing the energy transition.

Hours

Full-time | Schedule may vary depending on site operational needs; flexibility required

Responsibilities will include:

- Develop, test, and maintain production software systems powering automated battery sorting, spanning ML inference, image acquisition, sensor integration, and hardware-adjacent control interfaces
- Train and deploy computer vision models for battery chemistry classification, including dataset annotation, preprocessing, and evaluation within established data pipelines

- Build and maintain services and APIs that connect ML outputs to downstream systems including MES, HMI, and PLC/controls interfaces
- Own observability across the production software stack through structured logging, metrics dashboards, alerting, and on-call triage for inference pipelines and supporting services
- Monitor model performance in production to catch regressions or distribution shifts and drive iterative improvements through data analysis and retraining
- Contribute to infrastructure-as-code and CI/CD workflows to validate, version, and deploy application code and ML model artifacts to production environments
- Collaborate cross-functionally with Controls, Hardware, Manufacturing, DevOps, and IT teams to translate operational needs into software and model improvements

Desired Qualifications:

- B.S. in Computer Science, Electrical Engineering, or a related field, or equivalent practical experience
- 2+ years of industry experience working with machine learning models, preferably in computer vision
- Hands-on experience with ML frameworks and libraries such as PyTorch and OpenCV
- Experience contributing to production codebases and pipelines with an emphasis on clean, well-documented, and well-tested code
- Experience designing and tracking ML experiments using tools such as MLflow
- Familiarity with edge deployment or model optimization techniques for inference (e.g., quantization, TensorRT, ONNX Runtime) in latency-sensitive or resource-constrained environments
- Experience with OCR, image classification pipelines, or multi-sensor and multimodal fusion
- Experience working in or alongside industrial, manufacturing, or operations environments where software interacts with physical systems
- Strong cross-functional communication skills and ability to prioritize and execute in a fast-paced, dynamic environment
- A passion for sustainability and making the world a better place!

Working Conditions:

- Factory floor environment; work schedule may vary depending on site operational needs and flexibility is required
- Willingness and ability to travel to Reno, NV as needed
- Additional working conditions to be confirmed with Hiring Manager

In accordance with California pay transparency laws, the salary range for this position is listed below. Actual compensation may vary based on a variety of factors, including experience, education, and skills.

California Pay Range:

$152,500—$200,000 USD

The position is full-time. Compensation will be commensurate with experience.

We collect personal information (PI) from you in connection with your application for employment with Redwood Materials, including the following categories of PI: identifiers, personal records, professional or employment information, and inferences drawn from your PI. We collect your PI for our purposes, including performing services and operations related to your potential employment. If you have additional privacy-related questions, please contact us at privacy@redwoodmaterials.com.

Archive notes for ECS 1100

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Degree / experience excerpt: B.S. in Computer Science, Electrical Engineering, or a related field, or equivalent practical experience | 2+ years of industry experience working with machine learning models, preferably in computer vision

Key skills listed in the archive: PyTorch; machine learning; computer vision; optimization

Archive metadata — Accessed October 26, 2026 (America/Chicago).

Recruiting status: Active · PDF: Complete · Source: Official employer Greenhouse Employer / ATS identifier: 6099577004

Career stage: Entry / Mid-career

Major fit: Direct skills pathway: data science, machine learning, analytics or data engineering; exact degree/experience requirements retained. Energy-sector connection: Computer vision for battery sorting or data engineering for energy-storage batteries.

Retrieval note: New posting found in expansion round. Official posting includes an application form with applicant input fields. Major fit is an archive classification; eligibility follows the employer qualifications.

Identity review: New posting: official employer title, location and identifier captured.. Source type: Official employer Greenhouse. Source identifier: 6099577004.

Sources and retrieval details

Employer posting

https://boards.greenhouse.io/redwoodmaterials/jobs/6099577004?gh_jid=6099577004

Retrieved source

https://boards-api.greenhouse.io/v1/boards/redwoodmaterials/jobs?content=true

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